United States Patent Application Publication	20250285260
Kind Code	A1
Publication Date	September 11, 2025
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METHOD FOR DETECTING SEMICONDUCTOR DEFECT USING TEMPERATURE DIFFERENCE CONTRAST

Abstract

A method for detecting semiconductor defect using temperature difference contrast is provided, including the following steps: obtaining an image of the first surface or a second surface of a standard semiconductor; heating part or all of a first surface of a target semiconductor by a heat source; stopping heating part or all of the first surface of the target semiconductor by the heat source; obtaining an image of the first surface or a second surface of the target semiconductor by a thermal imager; and comparing the first surface or the second surface of the standard semiconductor and the first surface or the second surface of the target semiconductor by a detecting unit to determine if the target semiconductor having defects or not. As such, the method can detect the target semiconductor having defects or not with temperature difference contrast under a condition of no damaging the target semiconductor.

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Family ID: **96949570**

Appl. No.: **18/613273**

Filed: **March 22, 2024**

Foreign Application Priority Data

TW	113108366	Mar. 07, 2024

Publication Classification

Int. Cl.: **G06T7/00** (20170101); **H05B1/02** (20060101); **H05B3/00** (20060101)

U.S. Cl.:

CPC **G06T7/001** (20130101); **H05B1/0233** (20130101); **H05B3/0047** (20130101); G06T2207/10048 (20130101); G06T2207/30148 (20130101)

Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present invention relates generally to a method for detecting semiconductor, and more particularly, to a method for detecting semiconductor defects using temperature difference contrast.

2. The Prior Arts

[0002] Semiconductor is a substance or material whose electrical conductivity is between that of metallic conductors and insulators. The current method for detecting defects in semiconductors is through sampling a batch of semiconductors, by destroying the sampled target semiconductors by slicing, and then using a scanning electron microscope to check the bonding accuracy to determine whether the target semiconductors are defective.

[0003] However, currently there is no detection method available to detect defects in the semiconductor without destroying the semiconductor.

SUMMARY OF THE INVENTION

[0004] A primary objective of the present invention is to provide a method for detecting semiconductor defects using temperature difference contrast, which can detect whether the target semiconductor has defects by using temperature difference contrast without damaging the target semiconductor.

[0005] In order to achieve the aforementioned objective, the present invention provides a method for detecting semiconductor defects using temperature difference contrast, which includes the following steps: obtaining an image of a first surface or a second surface of a standard semiconductor; applying a heat source to part or all of a first surface of a target semiconductor for the part or all of the first surface of the target semiconductor to absorb heat from the heat source; the heat source stopping heating part or all of the first surface of the target semiconductor, and part or all of the first surface of the target semiconductor completely diffusing heat in a direction towards part or all of a second surface of the target semiconductor; a thermal imager sensing temperatures of part or all of the first surface of the target semiconductor or temperatures of part or all of the second surface to obtain an image of the first surface or the second surface of the target semiconductor; and an inspection unit receiving the image of the first surface or the second surface of the standard semiconductor and the image of the first surface or the second surface of the target semiconductor, and comparing grayscale of the image of the first surface or the second surface of the standard semiconductor with the image of the first surface or the second surface of the target semiconductor to determine whether the target semiconductor having defects.

[0006] In a preferred embodiment, the step of obtaining an image of a first surface or a second surface of a standard semiconductor further includes: the heat source heating part or all of the first surface of the standard semiconductor, and the part or all of first surface of the standard semiconductor absorbing heat from the heat source; the heat source stopping heating part or all of the first surface of the standard semiconductor, and part or all of the first surface of the standard semiconductor diffusing heat in a direction towards part or all of the second surface of the standard semiconductor; and the thermal imager sensing the temperatures of part or all of the first surface of the standard semiconductor or the temperatures of part or all of the second surface to obtain an image of the first surface or the second surface of the standard semiconductor.

[0007] In a preferred embodiment, the step of sensing the temperatures of part or all of the first surface of the standard semiconductor or the temperatures of part or all of the second surface of the standard semiconductor by the thermal imager further includes: the thermal imager continuously capturing an image of the temperatures of part or all of the first surface of the standard semiconductor or an image of the temperatures of part or all of the second surface of the standard semiconductor to sense a change of the temperatures of part or all of the first surface of the standard semiconductor or a change of the temperatures of part or all of the second surface of the standard semiconductor.

[0008] In a preferred embodiment, the step of determining whether the target semiconductor having defects further includes: compared with the image of the first surface or the second surface of the standard semiconductor, the inspection unit detecting grayscale of the image of at least one point area of the first surface of the target semiconductor to be darker or grayscale of the image of at least one point area of the second surface of the target semiconductor to be lighter, so as to determine that the target semiconductor to be defective and the defect to be at least one conductive wire breakage.

[0009] In a preferred embodiment, the step of determining whether the target semiconductor having defects further includes: compared with the image of the first surface or the second surface of the standard semiconductor, the inspection unit detecting grayscale of the image of the first surface of the target semiconductor having at least one less point area or grayscale of the image of the second surface of the target semiconductor having at least one less point area, so as to determine that the target semiconductor to be defective and the defect to be at least one erroneous conductive wire or assembly misalignment.

[0010] In a preferred embodiment, the step of determining whether the target semiconductor having defects further includes: compared with the image of the first surface or the second surface of the standard semiconductor, the inspection unit detecting grayscale of the image of the first surface of the target semiconductor having at least one more point area or grayscale of the image of the second surface of the target semiconductor having at least one more point area, so as to determine that the target semiconductor to be defective and the defect to be at least one erroneous conductive wire or assembly misalignment.

[0011] In a preferred embodiment, the step of determining whether the target semiconductor is defective further includes: compared with the image of the first surface or the second surface of the standard semiconductor, the inspection unit detecting grayscale of the image of at least one block area of the first surface of the target semiconductor to be darker or grayscale of the image of at least one block area of the second surface of the target semiconductor to be lighter, so as to determine that the target semiconductor to be defective and the defect to be at least one non-conductive wire material to be damaged or erroneous material proportion.

[0012] In a preferred embodiment, the step of sensing temperatures of part or all of the first surface of the target semiconductor or temperatures of part or all of the second surface of the target semiconductor by the thermal imager further includes: the thermal imager continuously capturing the image of part or all of the first surface of the target semiconductor or the image of part or all of the second surface of the target semiconductor to sense a temperature change of part or all of the first surface of the target semiconductor or a temperature change of part or all of the second surface of the target semiconductor.

[0013] In a preferred embodiment, the heat source applies heat for less than 0.1 second at a temperature greater than 50° C.

[0014] In a preferred embodiment, the heat source is a surface light source or a point light source.

[0015] The effect of the present invention is that the method of the present invention can use the temperature difference contrast to detect whether the target semiconductor has defects without damaging the target semiconductor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] The present invention will be apparent to those skilled in the art by reading the following detailed description of a preferred embodiment thereof, with reference to the attached drawings, in which:

[0017] FIG. **1** is a flowchart of a first embodiment of the method of the present invention.

[0018] FIG. **2** is a schematic view of steps S**10** to S**30** of the first embodiment of the method of the present invention.

[0019] FIG. **3** is a schematic view of steps S**40** to S**60** of the first embodiment of the method of the present invention.

[0020] FIG. **4** is an image of the first surface of a standard die according to the first embodiment of the method of the present invention.

[0021] FIG. **5** is an image of the first surface of a target die according to the first embodiment of the method of the present invention.

[0022] FIG. **6** is a schematic view of the connection relationship between the thermal imager and the inspection unit of the present invention.

[0023] FIG. **7** is an image of the first surface of a standard die according to a second embodiment of the method of the present invention.

[0024] FIG. **8** is an image of the first surface of a target die according to the second embodiment of the method of the present invention.

[0025] FIG. **9** is an image of the first surface of a standard die according to a third embodiment of the method of the present invention.

[0026] FIG. **10** is an image of the first surface of a target die according to the third embodiment of the method of the present invention.

[0027] FIG. **11** is a schematic view of steps S**10** to S**30** of a fourth embodiment of the method of the present invention.

[0028] FIG. **12** is a schematic view of steps S**40** to S**60** of the fourth embodiment of the method of the present invention.

[0029] FIG. **13** is a flow chart of a fifth embodiment of the method of the present invention.

[0030] FIG. **14** is a schematic view of the first cycle of step S**10** to step S**30** in the fifth embodiment of the method of the present invention.

[0031] FIG. **15** is a schematic view of the first cycle of step S**40** to step S**60** of the fifth embodiment of the method of the present invention.

[0032] FIG. **16** is a schematic view of the second cycle of step S**10** to step S**30** in the fifth embodiment of the method of the present invention.

[0033] FIG. **17** is a schematic view of the second cycle of step S**40** to step S**60** of the fifth embodiment of the method of the present invention.

[0034] FIG. **18** is a schematic view of the third cycle of step S**10** to step S**30** in the fifth embodiment of the method of the present invention.

[0035] FIG. **19** is a schematic view of the third cycle of step S**40** to step S**60** in the fifth embodiment of the method of the present invention.

[0036] FIG. **20** is a schematic view of the fourth cycle of step S**10** to step S**30** of the fifth embodiment of the method of the present invention.

[0037] FIG. **21** is a schematic view of the fourth cycle from step S**40** to step S**60** of the fifth embodiment of the method of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0038] The accompanying drawings are included to provide a further understanding of the invention, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

[0039] The present invention provides a method for detecting semiconductor defects using temperature difference contrast, which includes the following steps:

[0040] Step S**10**, as shown in FIGS. **1** and **2**, a heat source **10** is a surface light source **11**, the surface light source **11** is an infrared heater, and the infrared heater faces first ends **211**, **221** of a plurality of conductive wires **21** and **22** of a standard die **20** (all of the first surface of the standard semiconductor) to projects infrared rays, the infrared rays heat the first ends **211**, **221** of the conductive wires **21**, **22**, and the first ends **211**, **221** of the conductive wires **21**, **22** absorb the heat of the infrared rays. Preferably, the infrared heater is a near-infrared heater or a short-wave far-infrared heater.

[0041] Step S**20**, as shown in FIGS. **1** and **2**, the infrared heater stops projecting infrared rays toward the first ends **211**, **221** of the conductive wires **21**, **22**, and the infrared heater stops projecting infrared rays toward the first ends **211**, **221** of the conductive wires **21**, **22** so that the infrared heater stops heating the first ends **211**, **221** of the conductive wires **21**, **22**, and the heat of the first ends **211**, **221** of the conductive wires **21**, **22** is diffused in the direction towards second ends **212**, **222** of the conductive wires **21**, **22** (all of the second surface of the standard semiconductor).

[0042] Step S**30**, as shown in FIGS. **1**, **2** and **4**, a thermal imager **30** senses the temperatures of the first ends **211**, **221** of the conductive wires **21**, **22** to obtain an image of the first surface of the standard die **20**.

[0043] Step S**40**, as shown in FIGS. **1** and **3**, the infrared heater projects infrared rays toward first ends **411**, **421** of a plurality of conductive wires **41**, **42** of a target die **40** (all of the first surface of the target semiconductor). The first ends **411**, **421** of the conductive wires **41**, **42** are heated, and the first ends **411**, **421** of the conductive wires **41**, **42** absorb the heat of infrared rays.

[0044] Step S**50**, as shown in FIGS. **1** and **3**, the infrared heater stops projecting infrared rays toward the first ends **411**, **421** of the conductors **41**, **42** so that the infrared heater stops heating the first ends **411**, **421** of the conductors **41**, **42**, and the heat of the first ends **411**, **421** of the conductive wires **41**, **42** is diffused in the direction towards the second ends **412**, **422** of the conductive wires **41**, **42** (all of the second surface of the target semiconductor).

[0045] In step S**60**, as shown in FIGS. **1**, **3** and **5**, the thermal imager **30** senses the temperatures of the first ends **411**, **421** of the conductive wires **41**, **42** to obtain an image of the first surface of the target die **40**.

[0046] Step S**70**, as shown in FIGS. **1**, **4**, **5** and **6**, an inspection unit **50** receives the image of the first surface of the standard die **20** and the image of the first surface of the target die **40**, and compares grayscales of the image of the first surface of the standard die **20** with grayscales of the image of the first surface of the target die **40**; and, compared with the image of the first surface of the standard die **20**, the inspection unit **50** detects grayscales of the image of the first end **421** of the conductive wire **42** (a point area on the first surface of the target semiconductor) is darker, which means that the temperature of the first end **421** of the conductive wire **42** is higher than the temperature of the first end **411** of the remaining conductive wire **41** to determine that the target die **40** is defective and the defect is that the conductive wire **42** is broken.

[0047] Preferably, step S**30** further includes: the thermal imager **30** continuously captures images of the first ends **211**, **221** of the conductive wires **21**, **22** to sense the temperature change of the first ends **211**, **221** of the conductive wires **21**, **22**.

[0048] Preferably, step S**60** further includes: the thermal imager **30** continuously captures images of the first ends **411**, **421** of the conductive wires **41**, **42** to sense the temperatures change of the first ends **411**, **421** of the conductive wires **41**, **42**.

[0049] Preferably, the thermal imager **30** is an infrared thermal imager.

[0050] Preferably, the heating time of the heat source **10** is less than 0.1 seconds, and the heating temperature is higher than 50° C. In other words, the heat source **10** is instantly energized to a high temperature in a short period of time to achieve the detection operation.

[0051] The difference between the second embodiment and the first embodiment is that in step S**70**, as shown in FIGS. **7** and **8**, compared with the image of the first surface of the standard die **20**A, the inspection unit **50** detects that the grayscales of the image of first surface of the target die **40**A has one less point area **23** and has one more point area **43**. As such, the inspection unit **50** determines that the target die **40**A is defective and the defect is caused by incorrect wiring or assembly misalignment.

[0052] The difference between the third embodiment and the first embodiment is that in step S**70**, as shown in FIGS. **9** and **10**, compared with the image of the first surface of the standard die **20**B, the inspection unit **50** detects that the grayscales of the image the first surface of the target die **40**B has a block area **44** that is darker to determine that the target die **40**B is defective and the defect is that the non-conductive material is damaged or incorrect material ratio.

[0053] The difference between the fourth embodiment and the first embodiment is that: first, as shown in FIG. **11**, the thermal imager **30** senses the temperature of the second ends **212**, **222** of the conductive wires **21**, **22** to obtain the image of the second surface of the standard die **20**; second, as shown in FIG. **12**, the thermal imager **30** senses the temperature of the second ends **412**, **422** of the conductive wires **41**, **42** to obtain an image of the second surface of the target die **40**.

[0054] Preferably, step S**30** further includes: the thermal imager **30** continuously captures images of the second ends **212**, **222** of the conductive wires **21**, **22** to sense the temperature change of the second ends **212**, **222** of the conductive wires **21**, **22**.

[0055] Preferably, step S**60** further includes: the thermal imager **30** continuously captures images of the second ends **412**, **422** of the conductive wires **41**, **42** to sense the temperature change of the second ends **412**, **422** of the conductive wires **41**, **42**.

[0056] In some embodiments, the surface light source **11** is a laser heater.

[0057] The differences between the fifth embodiment and the first embodiment are as follows:

[0058] In the first cycle, as shown in FIGS. **13**, **14** and **15**, step S**10**, the heat source **10** is a point light source **12**. The point light source **12** includes an infrared heater **121** and a convex lens **122**. The infrared heater **121** projects infrared rays towards the convex lens **122**, and the convex lens **122** focuses the infrared rays and irradiates the focused infrared rays on the first end **211** of one of the conductive wires **21** (part of the first surface of the standard semiconductor), and the infrared rays heat the first end **211** of one of the conductive wires **21**, and the first end **211** of one of the conductive wires **21** absorbs the heat of infrared rays; step S**20**, the infrared heater **121** stops projecting infrared rays toward the convex lens **122**, and the convex lens **122** stops focusing the infrared rays to irradiate on the first end **211** of one of the conductive wires **21**, i.e., the infrared heater **121** and the convex lens **122** stop heating the first end **211** of one of the conductive wires **21**, and the heat of the first end **211** of one of the conductive wires **21** diffuses toward the direction of the second end **212** of one of the conductive wires **21** (part of the second surface of the standard semiconductor); in step S**30**, the thermal imager **30** senses the temperature of the first end **211** of one of the conductive wires **21**; Step S**40**, the infrared heater **121** projects infrared rays toward the convex lens **122**, and the convex lens **122** focuses the infrared rays and irradiates the focused infrared rays on the first end **411** (part of the first surface of the target semiconductor) of one of the conductive wires **41**; the infrared rays heat the first end **411** of one of the conductive wires **41**, and the first end **411** of one of the conductive wires **41** absorbs the heat of infrared rays; in step S**50**, the infrared heater **121** stops projecting infrared rays toward the convex lens **122**, and the convex lens **122** stops focusing and irradiating on the first end **411** of one of the conductive wires **41**, i.e., the infrared heater **121** and the convex lens **122** stop heating the first end **411** of one of the conductive wires **41**, and the heat of the first end **41** of one of the conductive wires **41** diffuses towards the direction of the second end **412** (part of the second surface of the target semiconductor) of one of the conductive wires **41**; and in step S**60**, the thermal imager **30** senses the temperatures of the first end **211** of one of the conductive wires **21**.

[0059] In the second cycle, as shown in FIGS. **13**, **16** and **17**, the difference from the first cycle is that in step S**10**, the convex lens **122** focuses the infrared rays and irradiates the focused infrared rays on the first end **221** of the wire **22** (part of the first surface of the standard semiconductor), the infrared rays heat the first end **221** of the conductive wire **22**, and the first end **221** of the conductive wire **22** absorbs the heat of the infrared rays; in step S**20**, the convex lens **122** stops focusing the infrared ray to irradiate on the first end **221** of the conductive wire **22** so that the infrared heater **121** and the convex lens **122** stop heating the first end **221** of the conductive wire **22**, and the heat of the first end **221** of the conductive wire **22** diffuses toward the second end **222** of the conductive wire **22** (part of the second surface of the standard semiconductor); in step S**30**, the thermal imager **30** senses the temperature of the first end **221** of the conductive wire **22**; in step S**40**, the convex lens **122** focuses the infrared rays and irradiates on the first end **421** of the conductive wire **42** (part of the first surface of the target semiconductor), the infrared ray heats the first end **421** of the conductive wire **42**, and the first end **421** of the conductive wire **42** absorbs the heat of the infrared ray; step S**50**, the convex lens **122** stops focusing the infrared ray to irradiate on the first end **421** of the conductive wire **42** so that the infrared heater **121** and the convex lens **122** stops heating the first end **421** of the conductive wire **42**, and the heat of the first end **421** of the conductive wire **42** diffuses towards the direction of the second end **422** of the conductive wire **42** (part of the second surface of the target semiconductor); and in step S**60**, thermal imager **30** senses the temperature of first end **421** of the conductive wire **42**.

[0060] In the third cycle, as shown in FIGS. **13**, **18** and **19**, the difference from the first cycle is that in step S**10**, the convex lens **122** focuses the infrared rays and irradiates the focused infrared rays on the first end **211** of the other one of the conductive wires **21** (part of the first surface of a standard semiconductor), infrared rays heat the first end **211** of the other one of the conductive wires **21**, and the first end **211** of the other one of the conductive wires **21** absorbs the heat of the infrared ray; in step S**20**, the convex lens **122** stops focusing infrared rays to irradiate the first end **211** of the other one of the conductive wires **21** so that the infrared heater **121** and the convex lens **122** stop heating the first end **211** of the other one of the conductive wires **21**, and the heat of the first end **211** of the other of the conductive wires **21** diffuses towards the direction of the second end **212** of the other one of the conductive wires **21** (part of the second surface of the standard semiconductor); in step S**30**, thermal imager **30** senses the temperature of the first end **211** of the other one of the conductive wires **21**; in step S**40**, the convex lens **122** focuses the infrared ray and irradiates the focused infrared rays on the first end **411** of the other one of the conductive wires **41** (part of the first surface of the target semiconductor), the infrared ray heats the first end **411** of the other one of the conductive wires **41**, and the first end **411** of the other one of the conductive wires **41** absorbs the heat of the infrared ray; in step S**50**, the convex lens **122** stops focusing the infrared ray to irradiates on the first end **411** of the other one of the conductive wires **41**, so that the infrared heater **121** and the convex lens **122** stop heating the first end **411** of the other one of the conductive wires **41**, and the heat of the second end **411** of the other one of the conductive wires **41** diffuses towards the direction of the second end **412** of the other one of the conductive wires **41** (part of the second surface of the target semiconductor); and in step S**60**, the thermal imager **30** senses the temperature of the first end **411** of other one of the conductive wires **41**.

[0061] In the fourth cycle, as shown in FIGS. **13**, **20** and **21**, in step S**10**, the convex lens **122** focuses the infrared rays and irradiates the first end **211** of yet another one of the conductive wires **21** (part of the first surface of the standard semiconductor), the infrared ray heats the first end **211** of yet another one of the conductive wires **21**, and the first end **211** of the yet another one of the conductive wires **21** absorbs the heat of the infrared rays; in step S**20**, the convex lens **122** stops focusing the infrared ray irradiate on the first end **211** of the yet another one of the conductive wires **21**, so that the infrared heater **121** and the convex lens **122** stop heating the first end **211** of the yet another one of the conductive wires **21**, and the heat from the first end **211** of yet another one of the conductive wires **21** diffuses towards the direction of the second end **212** of yet another one of the conductive wires **21** (part of the second surface of the standard semiconductor); in step

S**30**, the thermal imager **30** senses the temperature of the first end **211** of yet another one of the conductive wires **21**; in step S**40**, the convex lens **122** focuses the infrared rays and irradiates the focused infrared rays on the first end **411** of yet another one of the conductive wires **41** (part of the first surface of the target semiconductor), and the infrared rays heat the first end **411** of yet another one of the conductive wires **41**, and the first end **411** of the yet another one of the conductive wires **41** absorbs the heat of the infrared rays; in step S**50**, the convex lens **122** stops focusing the infrared ray to irradiate on the first end **411** of yet another one of the conductive wires **41** so that the infrared heater **121** and the convex lens **122** stop heating the first end **411** of yet another one of the conductive wires **41**, and the heat of the first end **411** of yet another one of the conductive wires **41** diffuses towards the second end **412** of yet another one of the conductive wires **41** (part of the second surface of the target semiconductor); and in step S**60**, the thermal imager **30** senses the temperature of the first end **411** of yet another one of the conductive wires **41**.
[0062] After collecting the temperatures of the first ends **211**, **221** of the conductive wires **21**, **22** and the temperatures of the first ends **411**, **421** of the conductive wires **41**, **42**, the thermal imager **30** obtains an image of the first surface of the standard die **20** and an image of the first surface of the target die **40**. Finally, step S**70** is executed to determine that the target die **40** is defective and the defect is a breakage of conductive wire **22**.
[0063] In some embodiments, the point light source **12** is a laser heater.
[0064] In summary, the method of the present invention can detect whether the target semiconductor has defects by using the temperature difference contrast without damaging the target semiconductor.
[0065] Although the present invention has been described with reference to the preferred embodiments thereof, it is apparent to those skilled in the art that a variety of modifications and changes may be made without departing from the scope of the present invention which is intended to be defined by the appended claims.

Claims

1. A method for detecting semiconductor defects using temperature difference contrast, comprising the steps of: obtaining an image of a first surface or a second surface of a standard semiconductor; applying a heat source to part or all of a first surface of a target semiconductor for the part or all of the first surface of the target semiconductor to absorb heat from the heat source; the heat source stopping heating part or all of the first surface of the target semiconductor, and part or all of the first surface of the target semiconductor completely diffusing heat in a direction towards part or all of a second surface of the target semiconductor; a thermal imager sensing temperatures of part or all of the first surface of the target semiconductor or temperatures of part or all of the second surface to obtain an image of the first surface or the second surface of the target semiconductor; and an inspection unit receiving the image of the first surface or the second surface of the standard semiconductor and the image of the first surface or the second surface of the target semiconductor, and comparing grayscale of the image of the first surface or the second surface of the standard semiconductor with the image of the first surface or the second surface of the target semiconductor to determine whether the target semiconductor having defects.
2. The method for detecting semiconductor defects using temperature difference contrast according to claim 1, wherein the step of obtaining an image of a first surface or a second surface of a standard semiconductor further includes: the heat source heating part or all of the first surface of the standard semiconductor, and the part or all of first surface of the standard semiconductor absorbing heat from the heat source; the heat source stopping heating part or all of the first surface of the standard semiconductor, and part or all of the first surface of the standard semiconductor diffusing heat in a direction towards part or all of the second surface of the standard semiconductor; and the thermal imager sensing the temperatures of part or all of the first surface of the standard

semiconductor or the temperatures of part or all of the second surface to obtain an image of the first surface or the second surface of the standard semiconductor.

3. The method for detecting semiconductor defects using temperature difference contrast according to claim 2, wherein the step of sensing the temperatures of part or all of the first surface of the standard semiconductor or the temperatures of part or all of the second surface of the standard semiconductor by the thermal imager further includes: the thermal imager continuously capturing an image of the temperatures of part or all of the first surface of the standard semiconductor or an image of the temperatures of part or all of the second surface of the standard semiconductor to sense a change of the temperatures of part or all of the first surface of the standard semiconductor or a change of the temperatures of part or all of the second surface of the standard semiconductor.

4. The method for detecting semiconductor defects using temperature difference contrast according to claim 1, wherein the step of determining whether the target semiconductor having defects further includes: compared with the image of the first surface or the second surface of the standard semiconductor, the inspection unit detecting grayscale of the image of at least one point area of the first surface of the target semiconductor to be darker or grayscale of the image of at least one point area of the second surface of the target semiconductor to be lighter, so as to determine that the target semiconductor to be defective and the defect to be at least one conductive wire breakage.

5. The method for detecting semiconductor defects using temperature difference contrast according to claim 1, wherein the step of determining whether the target semiconductor having defects further includes: compared with the image of the first surface or the second surface of the standard semiconductor, the inspection unit detecting grayscale of the image of the first surface of the target semiconductor having at least one less point area or grayscale of the image of the second surface of the target semiconductor having at least one less point area, so as to determine that the target semiconductor to be defective and the defect to be at least one erroneous conductive wire or assembly misalignment.

6. The method for detecting semiconductor defects using temperature difference contrast according to claim 1, wherein the step of determining whether the target semiconductor having defects further includes: compared with the image of the first surface or the second surface of the standard semiconductor, the inspection unit detecting grayscale of the image of the first surface of the target semiconductor having at least one more point area or grayscale of the image of the second surface of the target semiconductor having at least one more point area, so as to determine that the target semiconductor to be defective and the defect to be at least one erroneous conductive wire or assembly misalignment.

7. The method for detecting semiconductor defects using temperature difference contrast according to claim 1, wherein the step of determining whether the target semiconductor is defective further includes: compared with the image of the first surface or the second surface of the standard semiconductor, the inspection unit detecting grayscale of the image of at least one block area of the first surface of the target semiconductor to be darker or grayscale of the image of at least one block area of the second surface of the target semiconductor to be lighter, so as to determine that the target semiconductor to be defective and the defect to be at least one non-conductive wire material to be damaged or erroneous material proportion.

8. The method for detecting semiconductor defects using temperature difference contrast according to claim 1, wherein the step of sensing temperatures of part or all of the first surface of the target semiconductor or temperatures of part or all of the second surface of the target semiconductor by the thermal imager further includes: the thermal imager continuously capturing the image of part or all of the first surface of the target semiconductor or the image of part or all of the second surface of the target semiconductor to sense a temperature change of part or all of the first surface of the target semiconductor or a temperature change of part or all of the second surface of the target semiconductor.

9. The method for detecting semiconductor defects using temperature difference contrast according to claim 1, wherein the heat source applies heat for less than 0.1 second at a temperature greater than 50° C.

10. The method for detecting semiconductor defects using temperature difference contrast according to claim 1, wherein the heat source is a surface light source or a point light source.